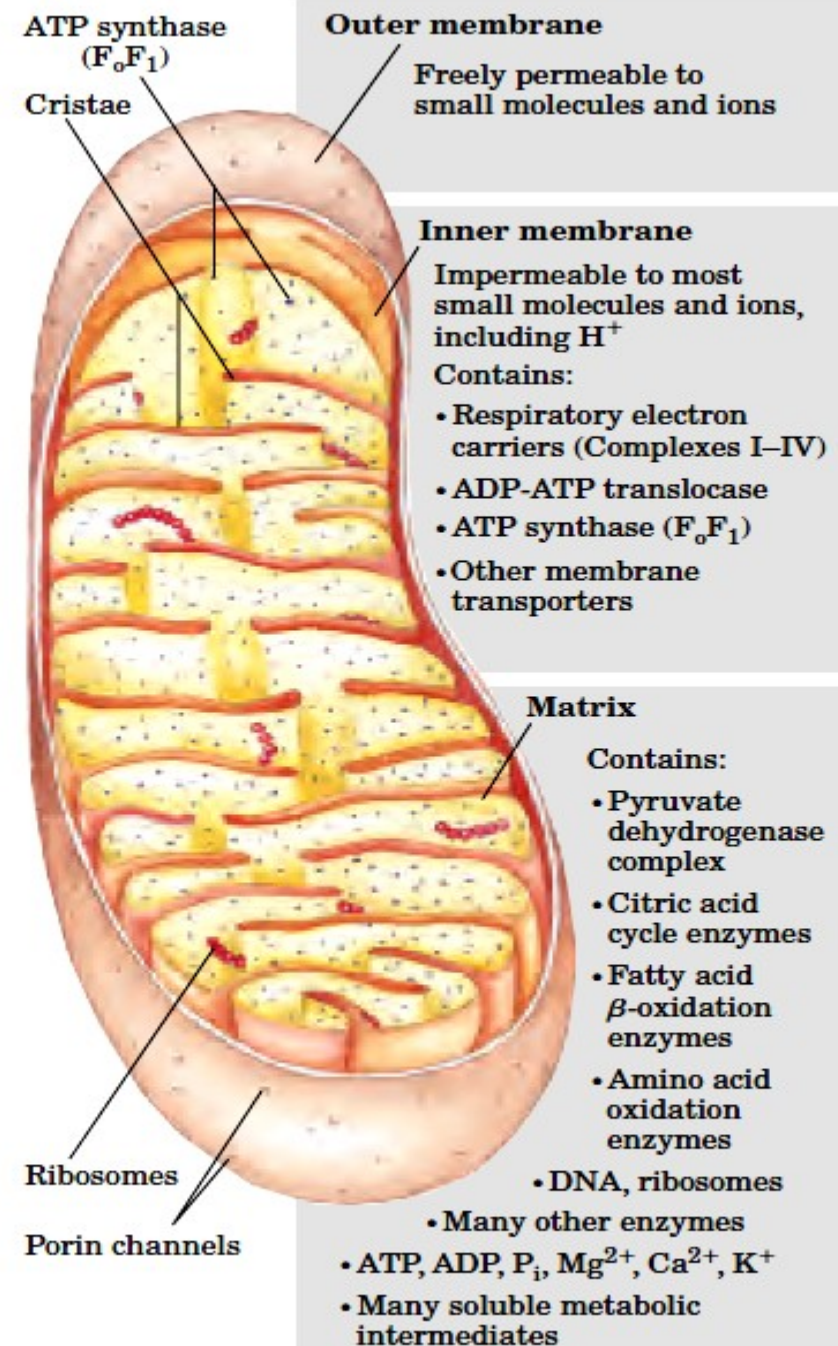


Electron Transfer Chain

Oxidative phosphorylation

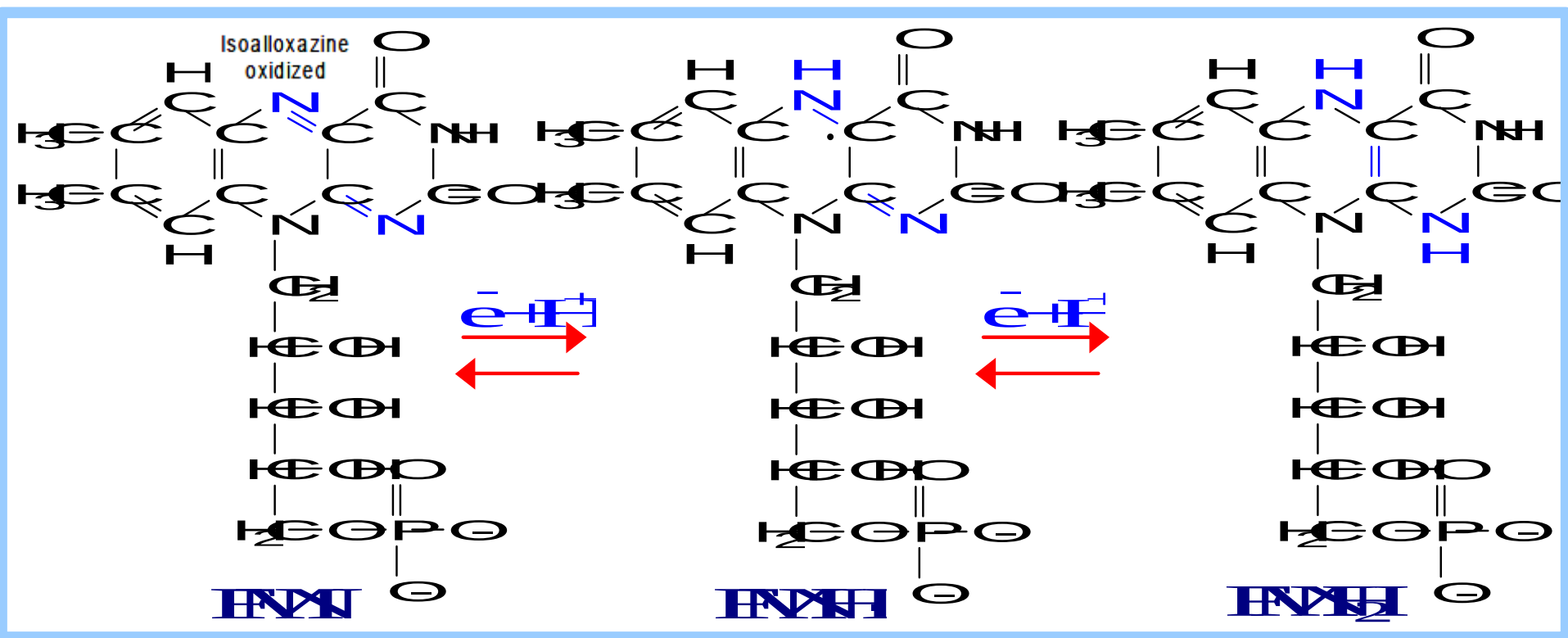
Mitochondria, has two membranes:

- ❖ The outer mitochondrial membrane is readily permeable to small molecules (*Mr* 5,000) and ions, which move freely through transmembrane channels formed by a family of integral membrane proteins called **porins**.
- ❖ The inner membrane is impermeable to most small molecules and ions, including protons (H^+).
- ❖ the only species that cross this membrane do so through specific transporters.
- ❖ The selectively permeable inner membrane segregates the intermediates and enzymes of cytosolic metabolic pathways from those of metabolic processes occurring in the matrix.



Electron Carriers

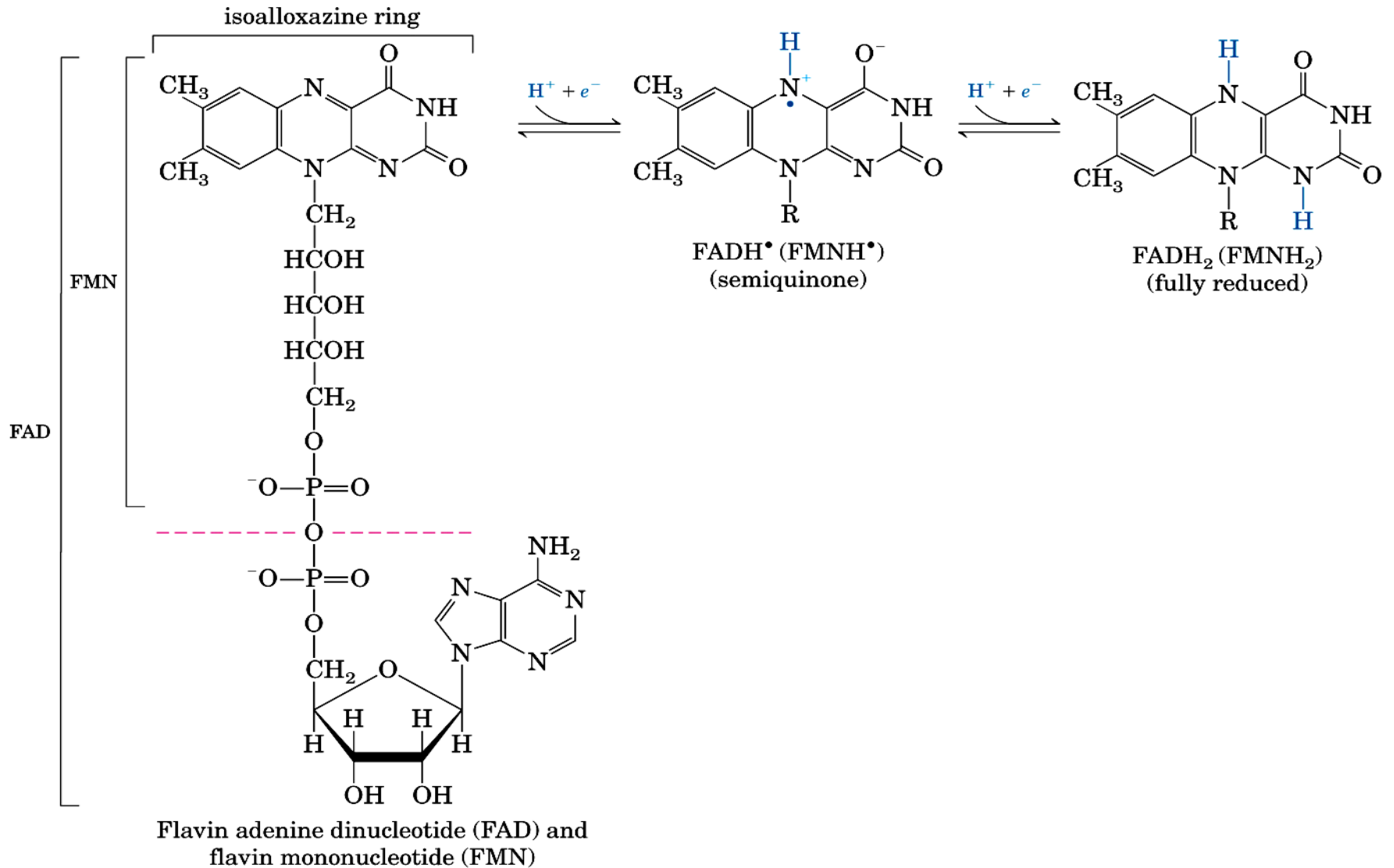
- ❖ Oxidative phosphorylation begins with the entry of electrons into the respiratory chain.
- ❖ Dehydrogenases collect electrons from catabolic pathways and funnel them into universal electron acceptors—nicotinamide nucleotides (NAD or NADP) or flavin nucleotides (FMN or FAD).
- ❖ **FMN** (**F**lavin **M**ono**N**ucleotide) is a prosthetic group of some **flavoproteins**.
- ❖ It is similar in structure to FAD (Flavin Adenine Dinucleotide), but lacking the adenine nucleotide.
- ❖ FMN (like FAD) can accept $2\text{ e}^- + 2\text{ H}^+$ to form **FMNH₂**.



FMN, when bound at the active site of some enzymes, can accept **1 e^-** to form the half-reduced **semiquinone radical**. The semiquinone can accept a 2nd e^- to yield FMNH_2 .

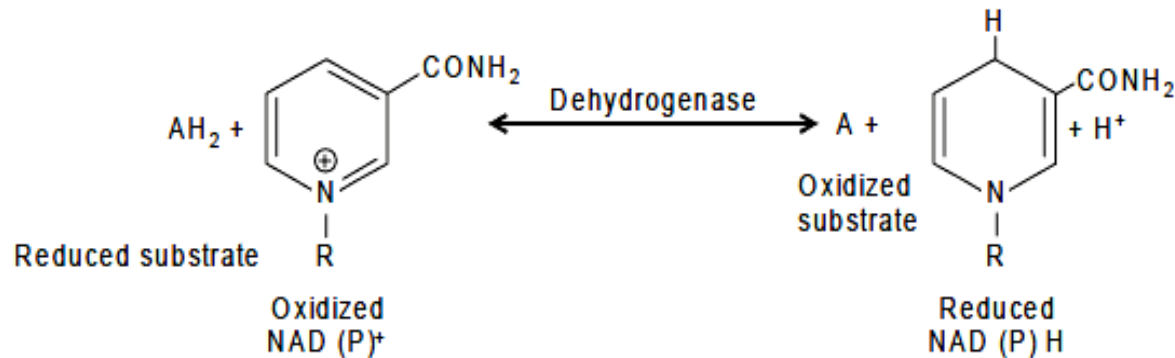
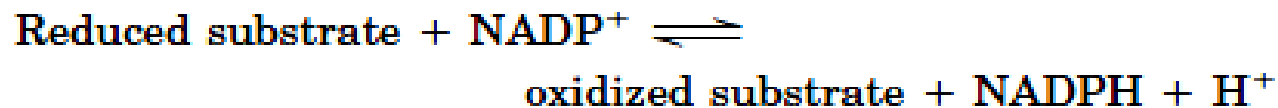
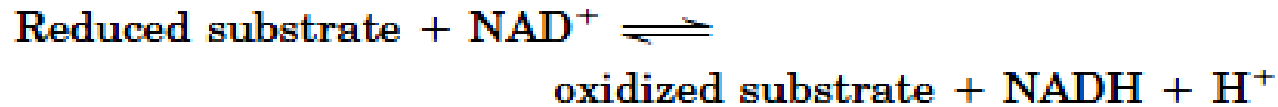
Since it can accept/donate **1 or 2 e^-** , FMN has an important **role** mediating e^- transfer between carriers that transfer 2e^- (e.g., NADH) & those that can accept only 1e^- (e.g., Fe^{+++}).

FAD



NAD(P)H

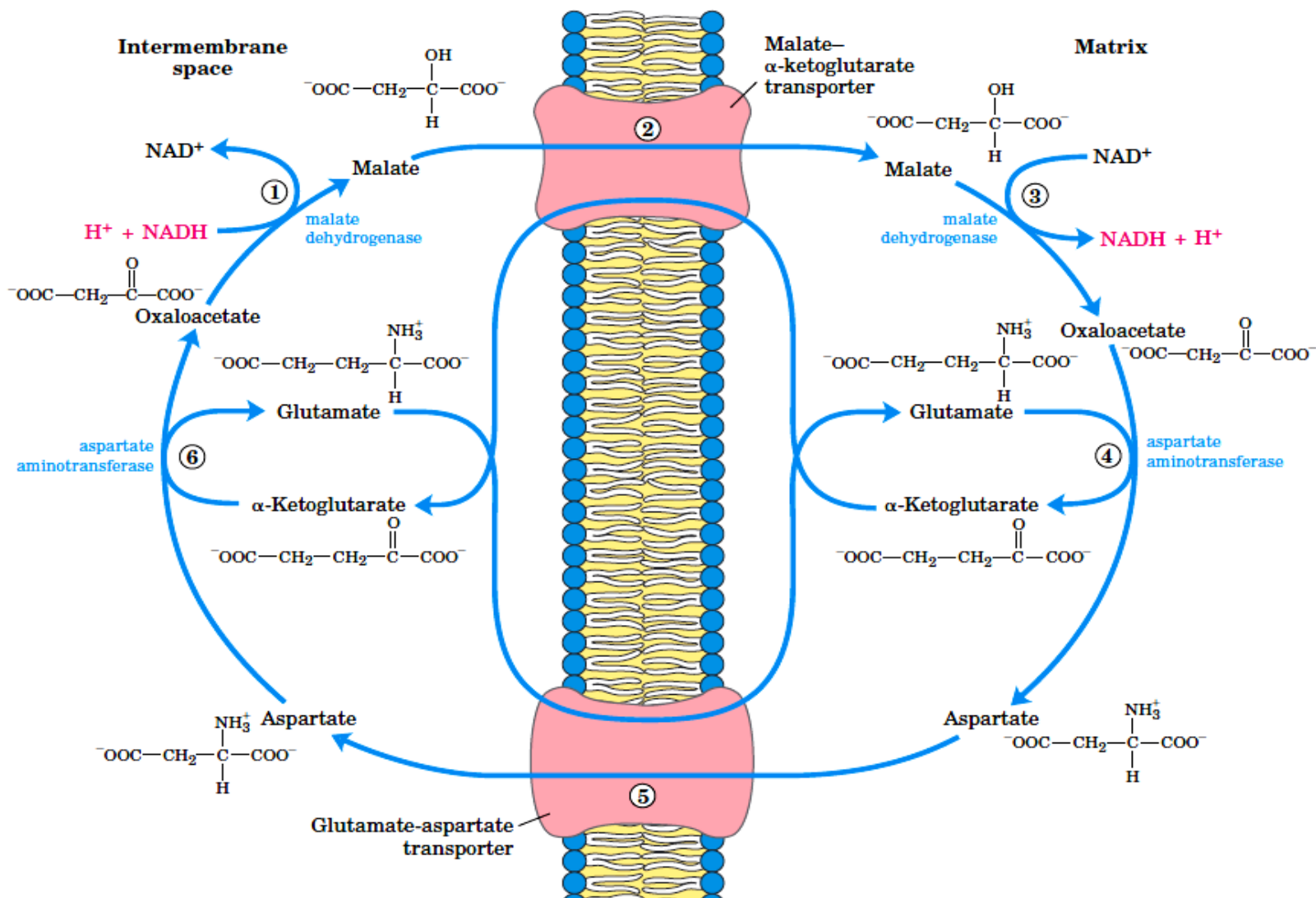
Nicotinamide nucleotide–linked dehydrogenases catalyze reversible reactions of the following general types:



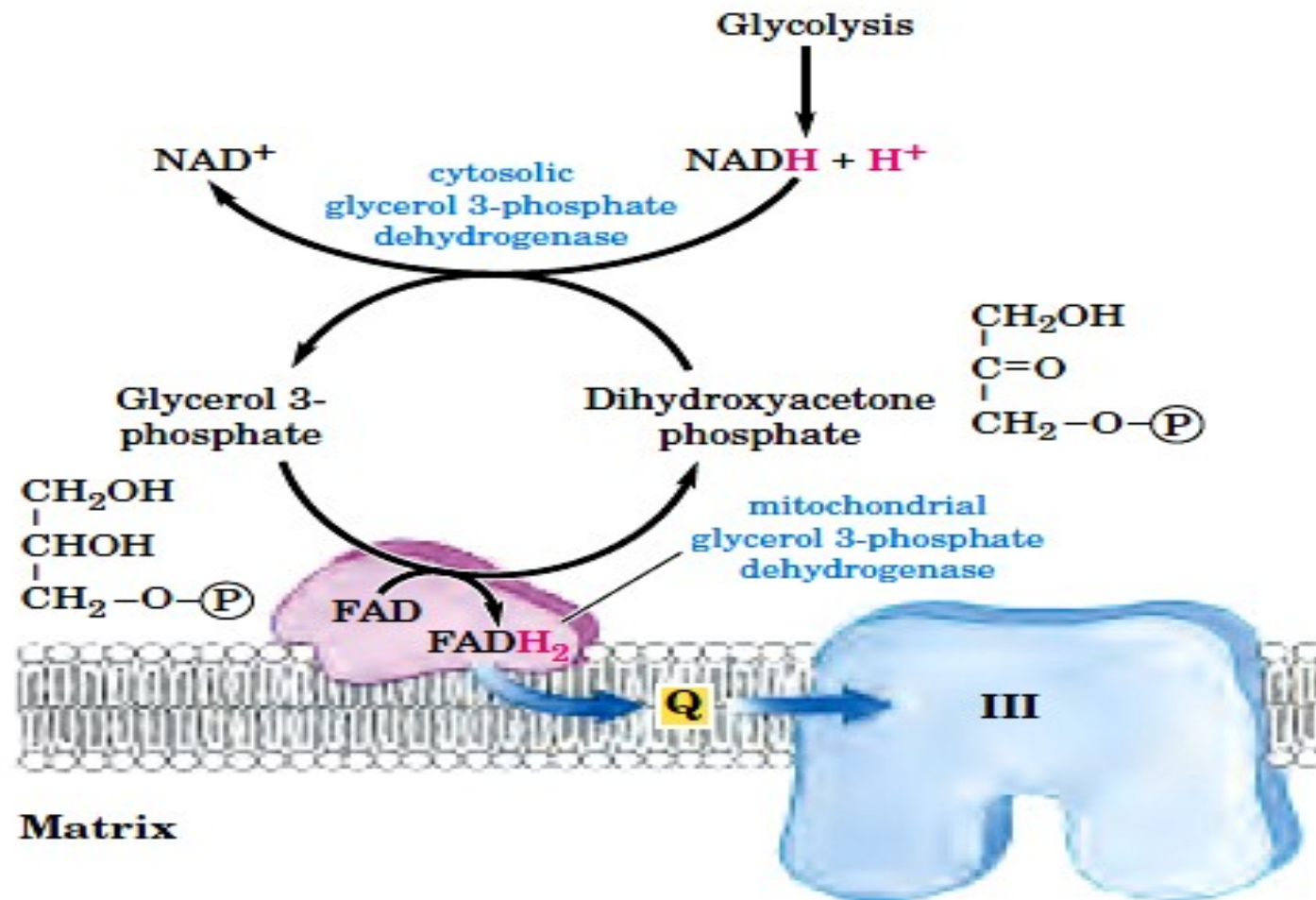
❖ Neither NADH nor NADPH can cross the inner mitochondrial membrane, but the electrons they carry can be shuttled across indirectly.

How?

Malate-aspartate shuttle. (liver, kidney, and heart mitochondria)



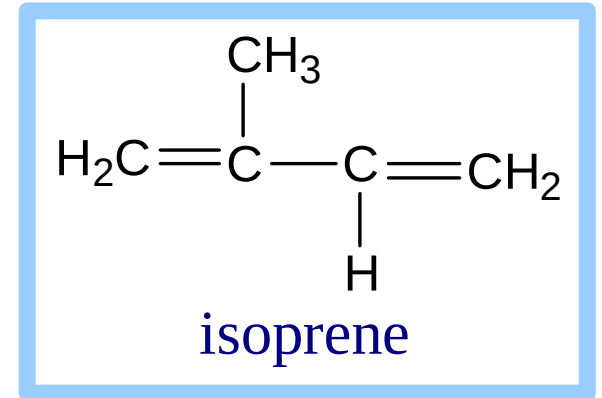
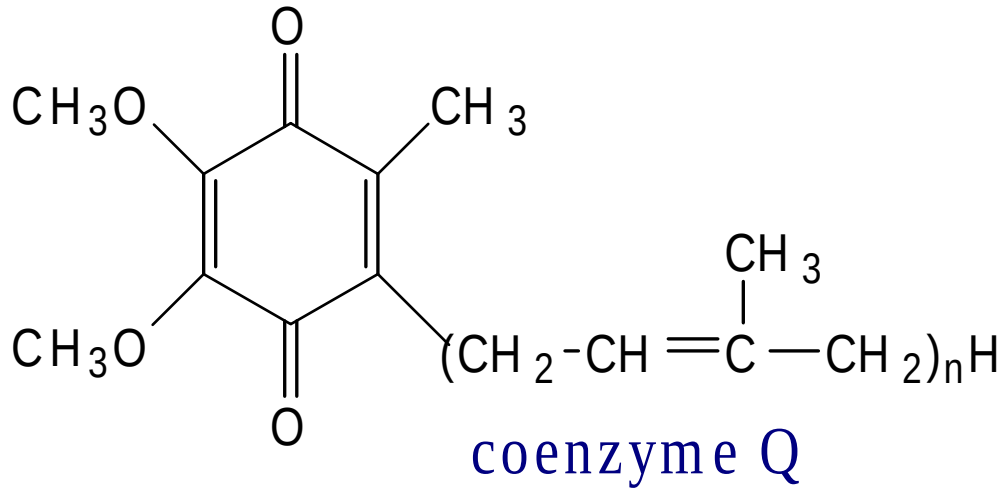
glycerol 3-phosphate shuttle (Skeletal muscle and brain)



In addition to NAD and flavoproteins, three other types of electron-carrying molecules function in the respiratory chain:

- ☐ Ubiquinone (Coenzyme Q).
- ☐ Cytochromes
- ☐ Iron sulfur proteins

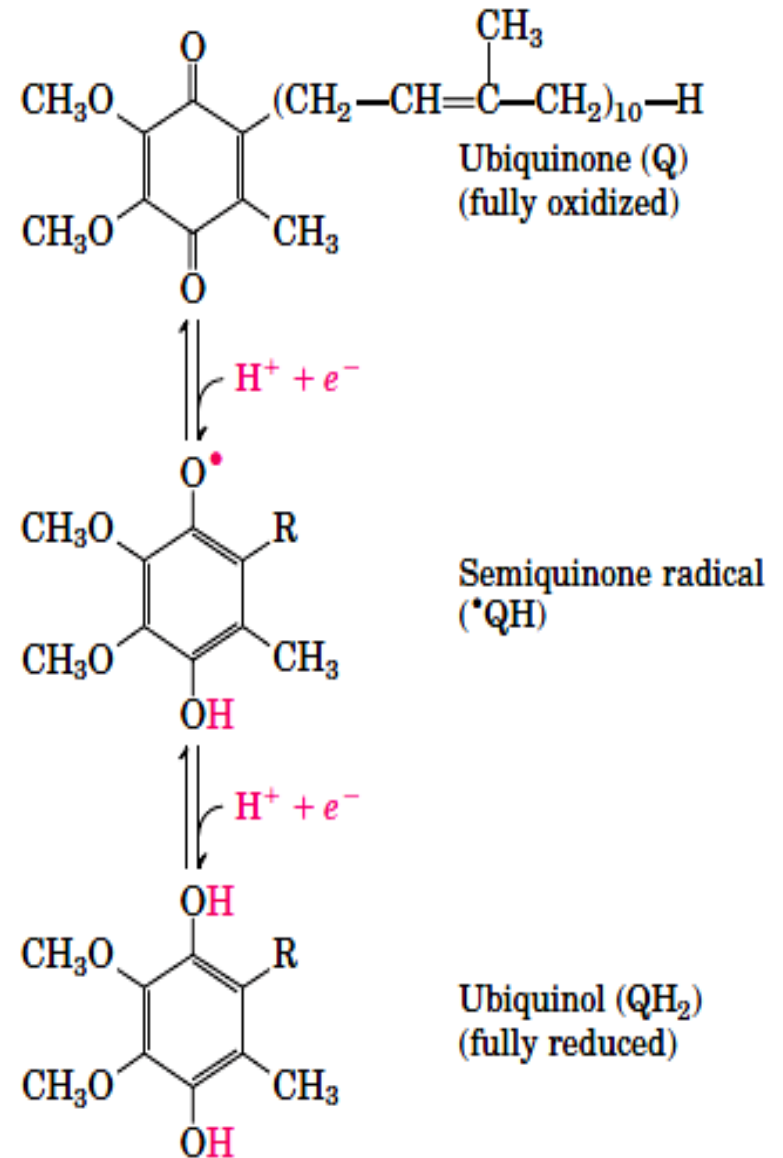
Coenzyme Q (Ubiquinone ,CoQ, Q)



- ❖ Is the only non-protein component of electron transport chain.
 - ❖ **Coenzyme Q** (CoQ, Q, ubiquinone) is very **hydrophobic**.
 - ✓ It dissolves in the hydrocarbon core of the inner mitochondrial membrane.
 - ❖ It includes a long **isoprenoid tail**, with multiple units having a carbon skeleton comparable to that of isoprene.
- In human cells, most often $n = 10$.

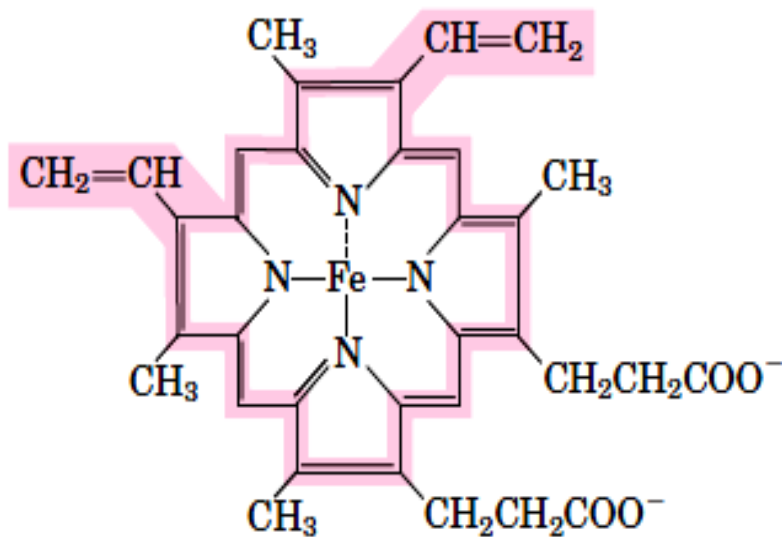
Ubiquinone can accept one electron to become the semiquinone radical ($\cdot\text{QH}$) or two electrons to form ubiquinol (QH_2) :

Coenzyme Q functions as a mobile e^- carrier within the mitochondrial inner membrane.



Cytochromes

- ❖ The **cytochromes** are **proteins with characteristic** strong absorption of visible light, due to their iron containing **heme prosthetic groups**.
- ❖ Three types exist: Cytochrome a, b and c.
- ❖ Are distinguished by differences in their light-absorption spectra.

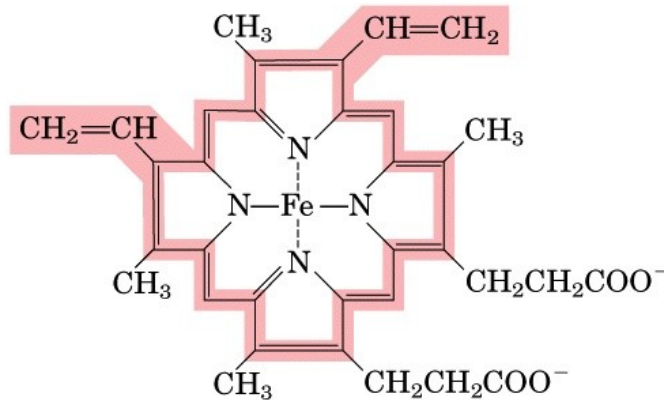


Iron protoporphyrin IX
(in b-type cytochromes)

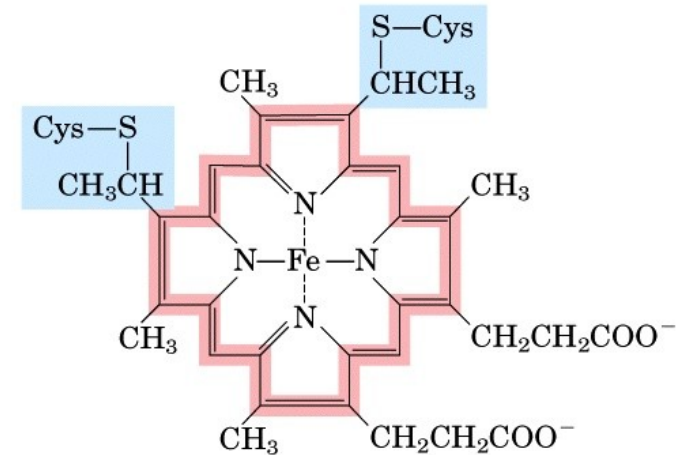
- ❖ The **Heme** prosthetic group of **cytochromes** contains an iron atom in a porphyrin ring system.
- ❖ The **Fe** is bonded to 4 **N** atoms of the porphyrin ring.
- ❖ The iron participates in redox reactions and oscillates between Fe²⁺ and Fe³⁺ states.
- ❖ Hemes in the 3 classes of cytochrome (**a**, **b**, **c**) differ slightly in substituents on the porphyrin ring system

Prosthetic groups of cytochromes

- Heme is a prosthetic group of **cytochromes**
- Mitochondria has 3 classes of cytochromes, designated *a*, *b*, and *c*

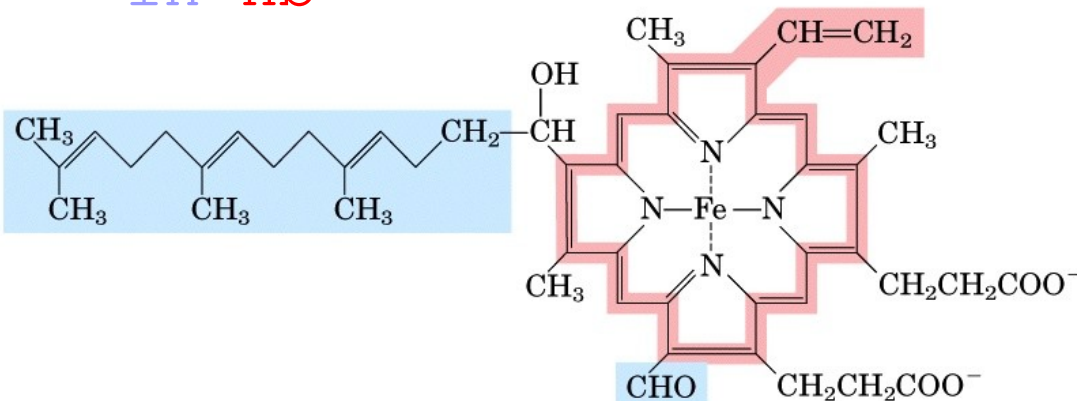


→ Iron protoporphyrin IX
(in *b*-type cytochromes)



Heme C
(in *c*-type cytochromes)

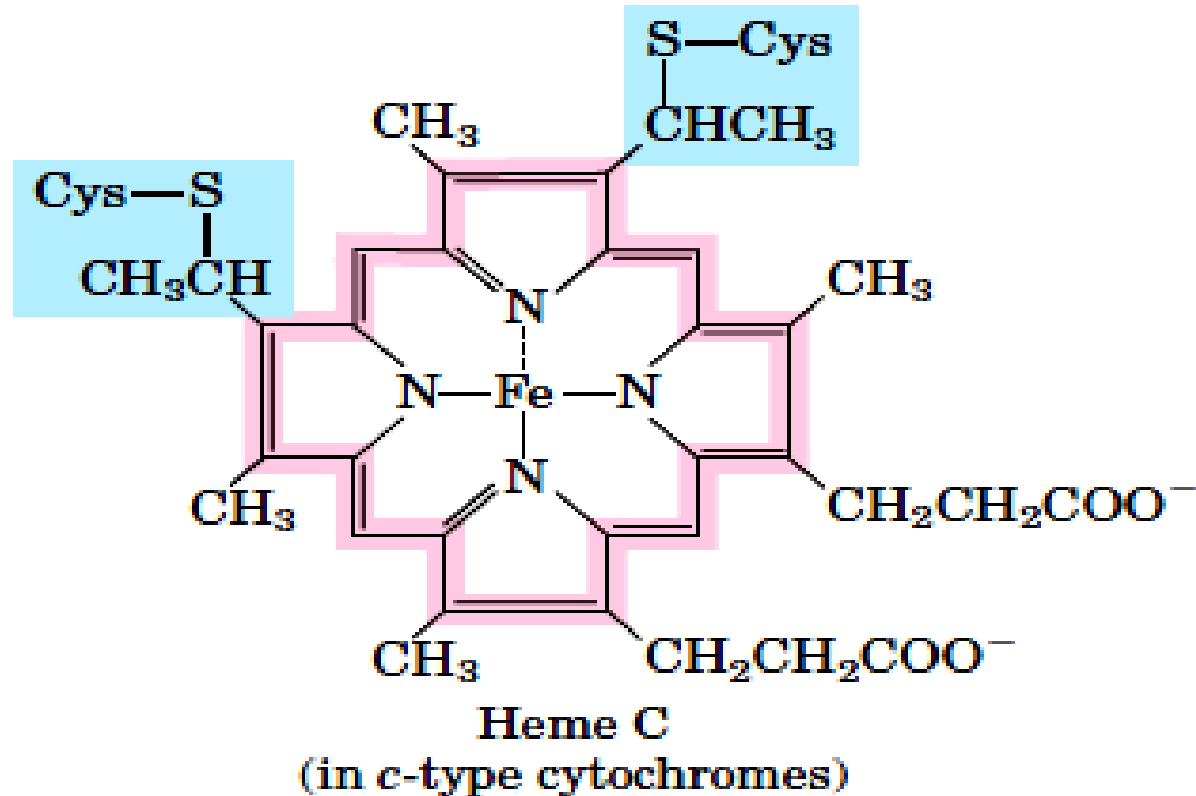
Found
in Hb



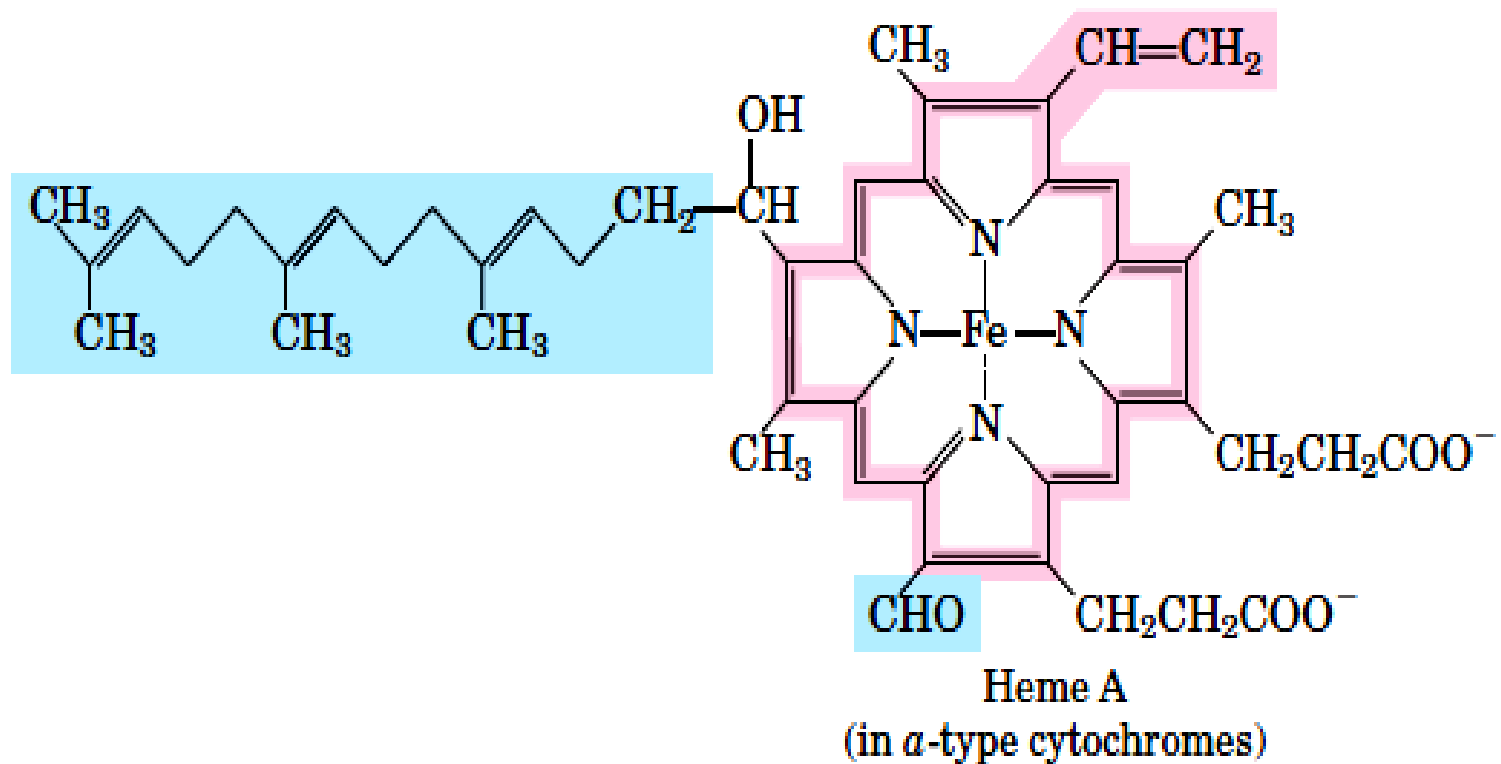
Heme A
(in *a*-type cytochromes)

The heme iron atom can undergo a 1 electron transition between ferric and ferrous states:





- ❖ Only **heme c** is covalently linked to the protein via thioether bonds to **cysteine** residues.



Heme a is unique in having a long farnesyl side-chain that includes 3 isoprenoid units.

Note: A common feature is 2 propionate side-chains.

Cytochromes are proteins with heme prosthetic groups. They absorb light at characteristic wavelengths.

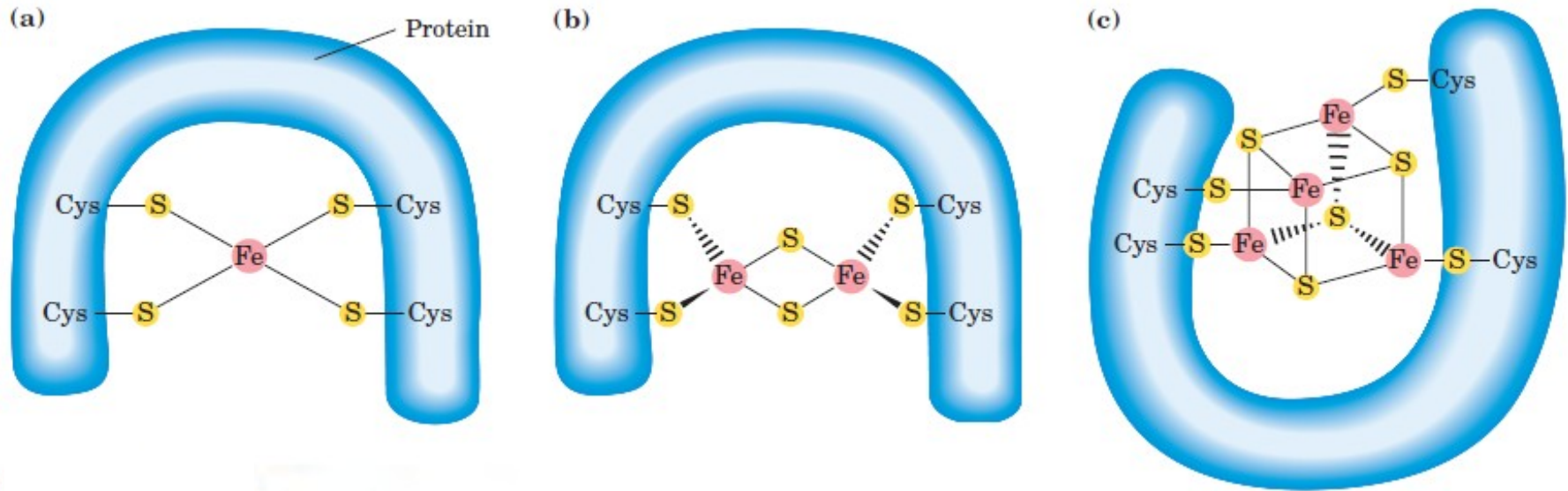
Absorbance changes upon oxidation/reduction of the heme iron provide a basis for monitoring heme redox state.

- ♦ Some cytochromes are part of large integral **membrane complexes**, each consisting of several polypeptides & including multiple electron carriers.

E.g., hemes a & a₃ that are part of the respiratory chain complex IV are often referred to as cytochromes a & a₃.

- ♦ **Cytochrome c** is instead a small, water-soluble protein with a single heme group.

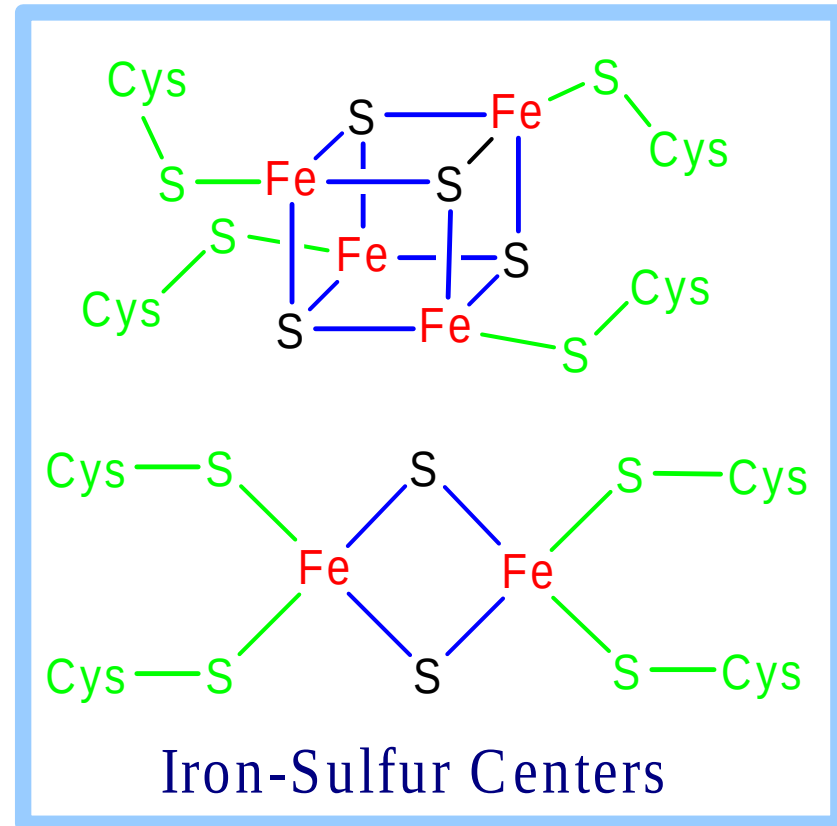
Iron-sulfur centers



Iron-sulfur centers (Fe-S) are prosthetic groups containing **2, 3, 4 or 8 iron atoms** complexed to elemental & cysteine **S**. **Cysteine** residues provide **S** ligands to the iron, while also holding these prosthetic groups in place within the protein.

Electron transfer proteins may contain multiple Fe-S centers.

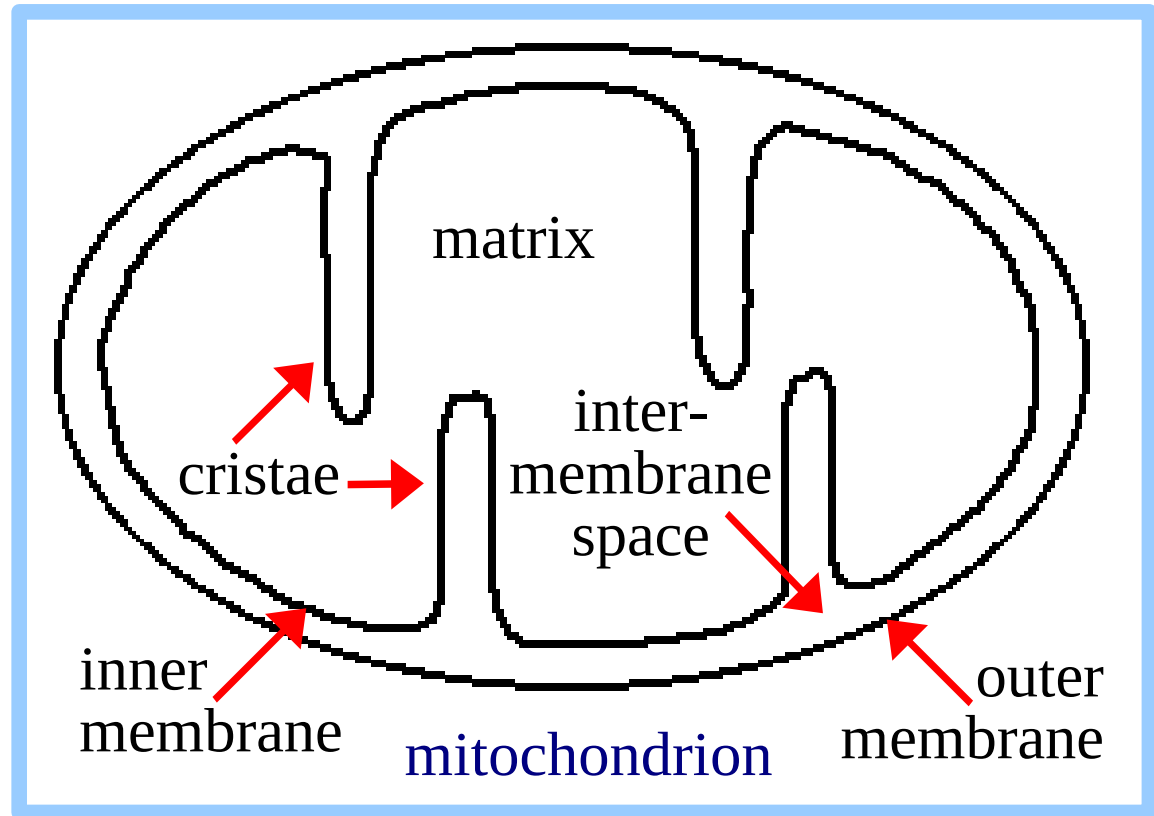
Iron-sulfur centers transfer only **one electron**, even if they contain two or more iron atoms, because of the close proximity of the iron atoms.



E.g., a 4-Fe center might cycle between redox states:



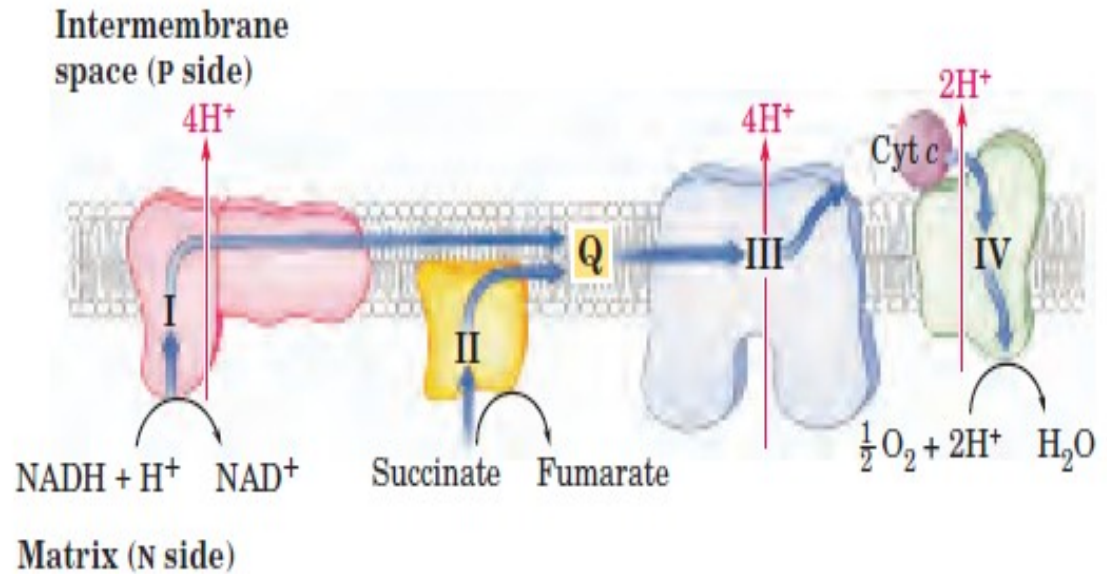
Respiratory Chain:



❖ Most constituents of the respiratory chain are embedded in the **inner mitochondrial membrane** (or in the cytoplasmic membrane of aerobic bacteria).

❖ The inner mitochondrial membrane has infoldings called **cristae** that increase the membrane area.

Electrons are transferred from **NADH** $\rightarrow$ **O₂** via multisubunit inner membrane **complexes I, II, III & IV**, plus **CoQ** & **cyt c**.

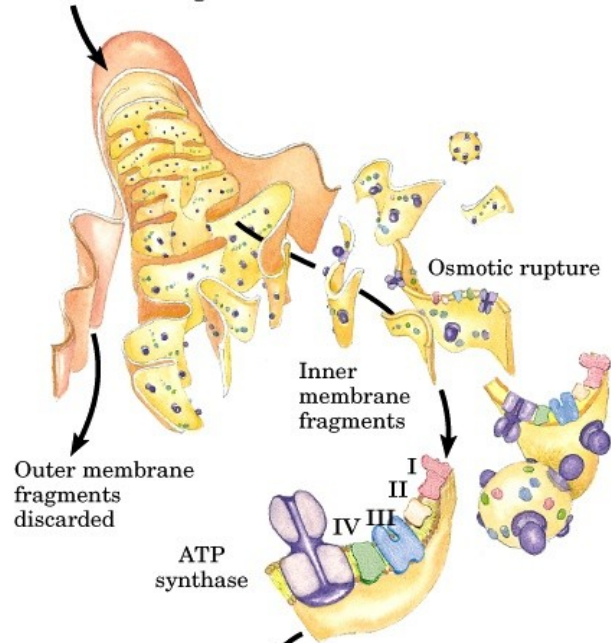


Within each complex, electrons pass sequentially through a series of electron carriers.

CoQ is located in the lipid core of the membrane. There are also binding sites for CoQ within protein complexes with which it interacts.

Cytochrome c resides in the intermembrane space. It alternately binds to complex III or IV during e⁻ transfer.

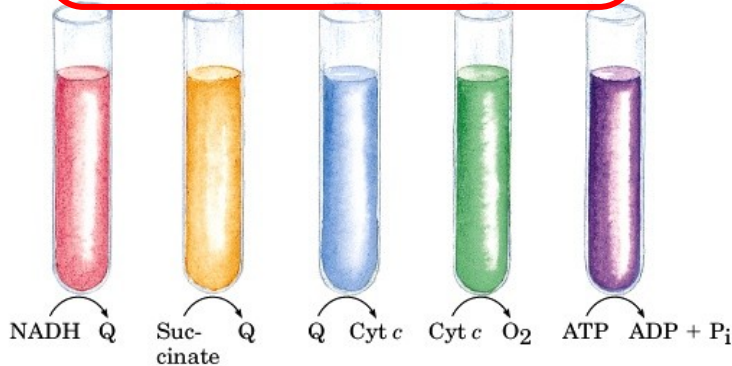
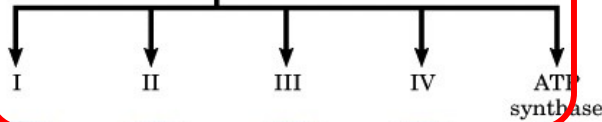
Treatment with digitonin



Individual **respiratory chain complexes** have been isolated and their composition determined.

Components of the electron transport chain can be purified from the mitochondrial inner membrane

Solubilization with detergent followed by ion-exchange chromatography

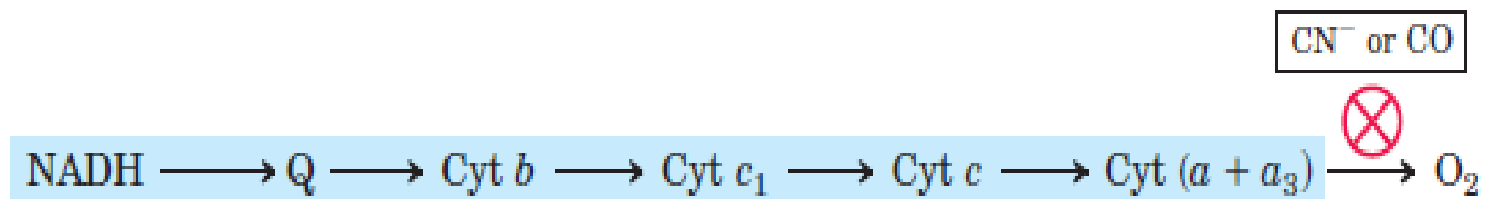
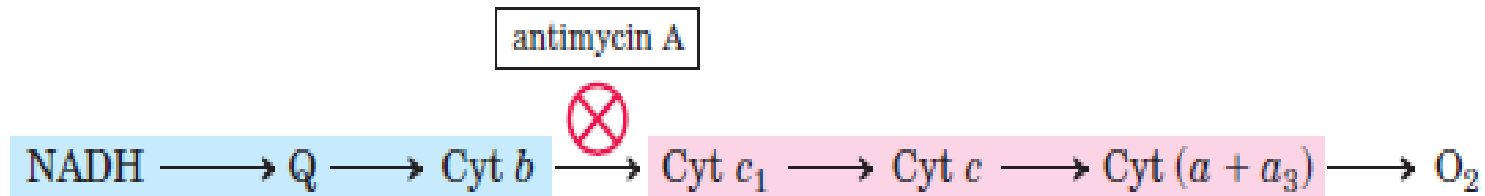
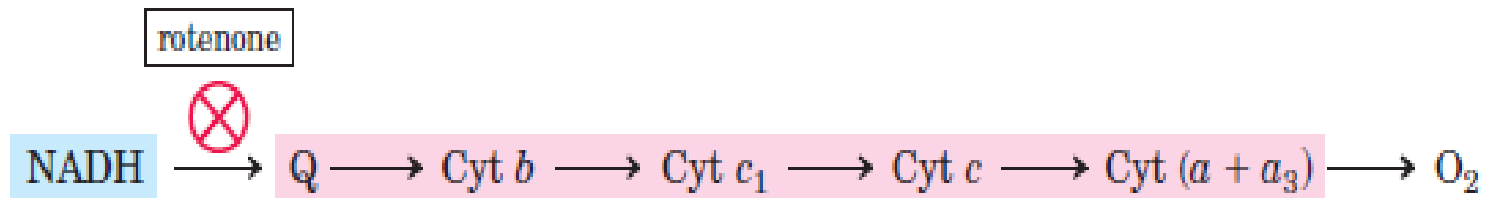


Reactions catalyzed by isolated fractions in vitro

There is also evidence for the existence of stable **supramolecular aggregates** containing multiple complexes.

Method for determining the sequence of electron carriers

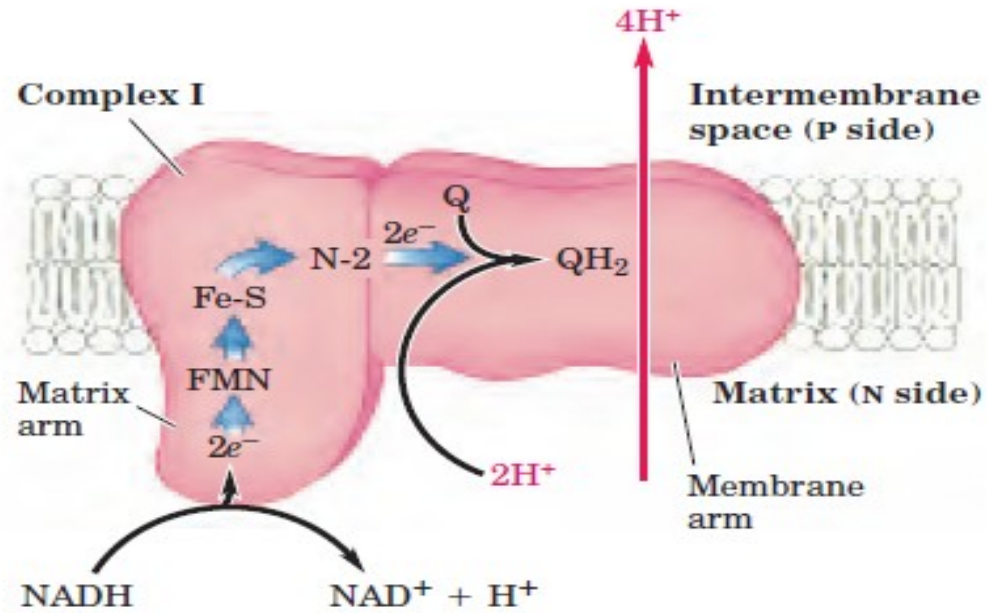
Inhibitors at various points of the respiratory chain were used



Composition of Respiratory Chain Complexes

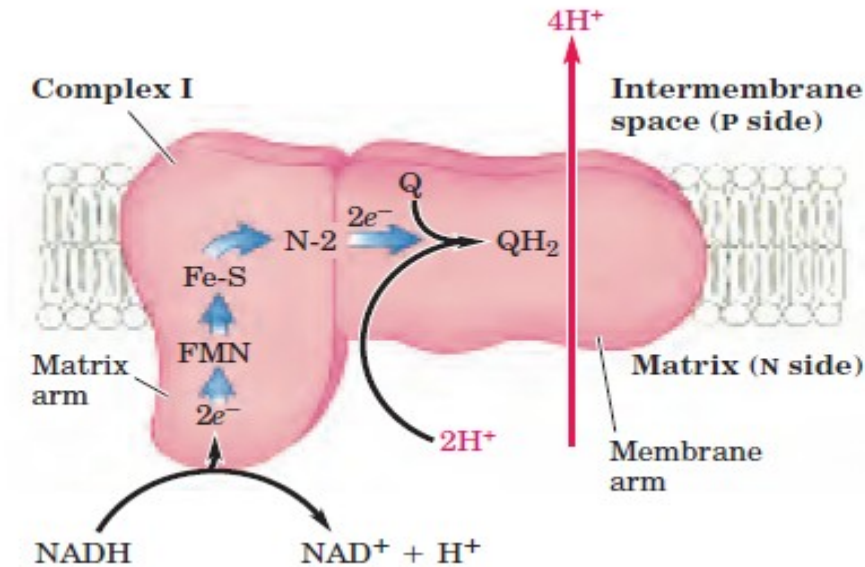
Complex	Name	No. of Proteins	Prosthetic Groups
Complex I	NADH Dehydrogenase	46	FMN, 9 Fe-S cntrs.
Complex II	Succinate-CoQ Reductase	5	FAD, cyt b ₅₆₀ , 3 Fe-S cntrs.
Complex III	CoQ-cyt c Reductase	11	cyt b _H , cyt b _L , cyt c ₁ , Fe-S _{Rieske}
Complex IV	Cytochrome Oxidase	13	cyt a, cyt a ₃ , Cu _A , Cu _B

Complex I
catalyzes
oxidation of
NADH, with
reduction of
coenzyme Q:



An atomic-level structure is not yet available for the entire complex I, which in mammals includes at least **46 proteins**, along with prosthetic groups **FMN** & several **Fe-S centers**.

Complex I is
L-shaped.



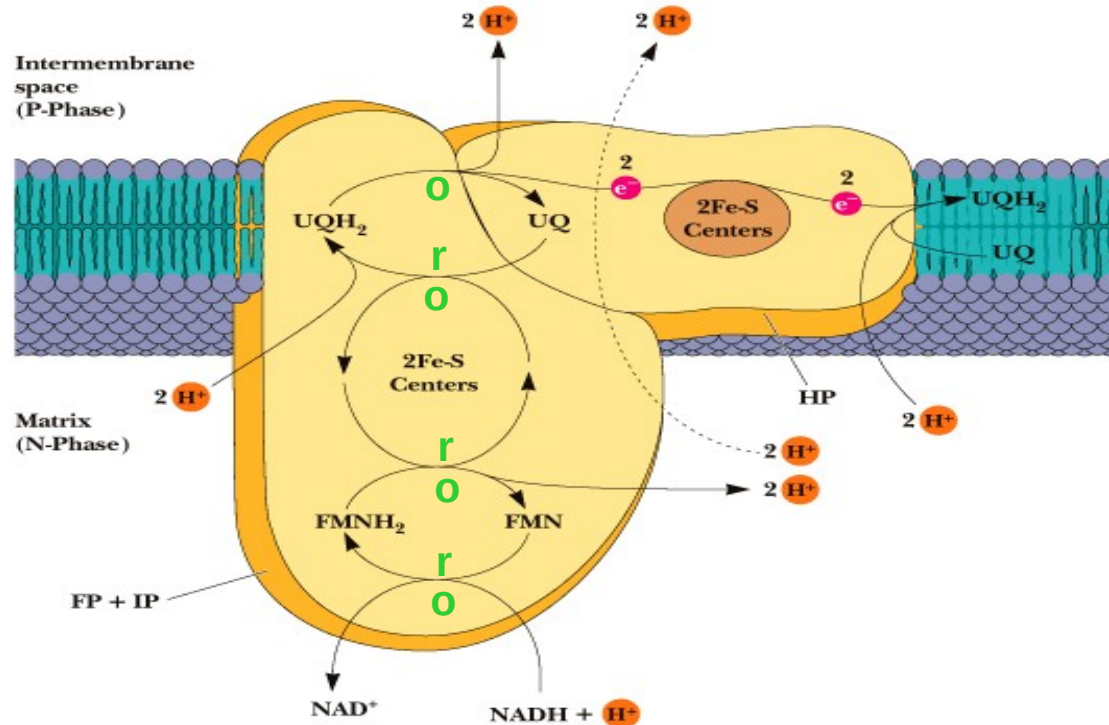
The **peripheral domain**, containing the **FMN** that accepts $2e^-$ from **NADH**, protrudes into the mitochondrial matrix.

Iron-sulfur centers are also located in the hydrophilic peripheral domain, where they form a **pathway for e^- transfer** from **FMN** to **coenzyme Q**.

A binding site for **coenzyme Q** is thought to be close to the interface between peripheral and intra-membrane domains.

Mechanism of e⁻ transfer in Complex I

Garrett & Grisham: Biochemistry, 2/e
Figure 21.6



Saunders College Publishing

- Estimated mass of this complex 850 kD
- Involves more than 30 polypeptide chains
- One molecule of FMN
- As many as 7 Fe-S clusters (2Fe-2S & 4Fe-4S)
- Precise mechanism of this complex is unknown

The initial electron transfers are:



After Fe-S is reoxidized by transfer of the electron to the next iron-sulfur center in the pathway:



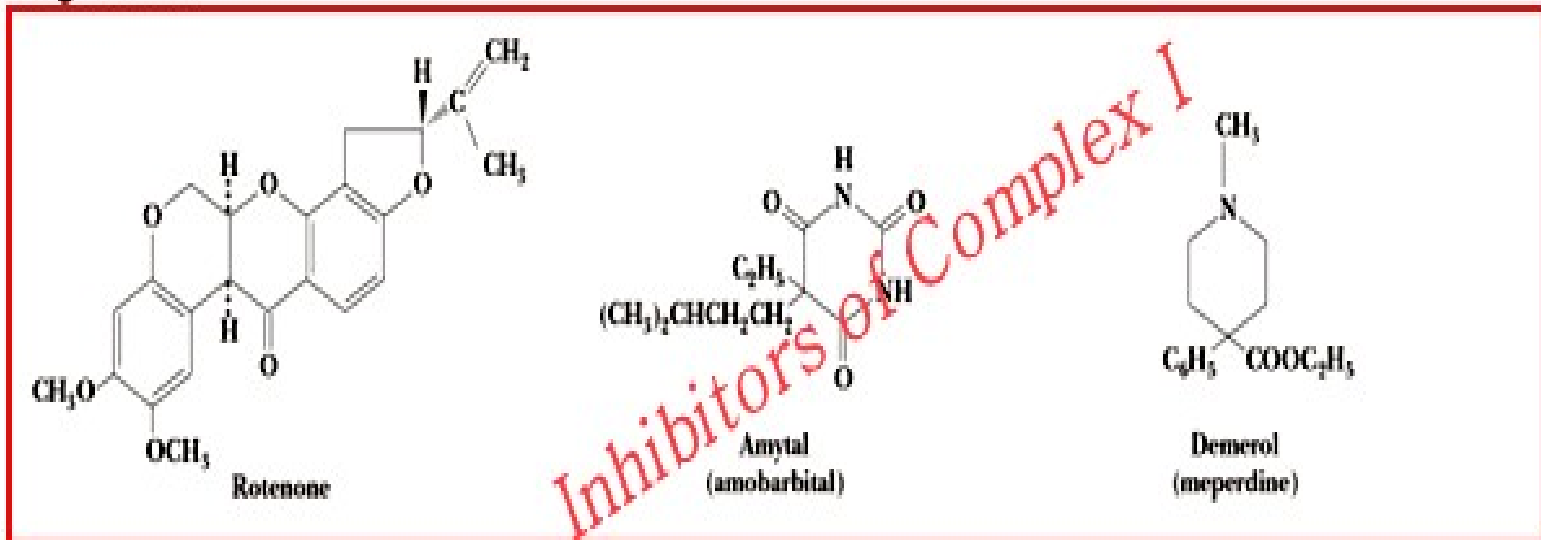
Electrons pass through a series of iron-sulfur centers, and are eventually transferred to coenzyme Q.

Coenzyme Q accepts 2 e⁻ and picks up 2 H⁺ to yield the fully reduced **QH₂**.

Inhibitors of complex 1

Amytal (a barbiturate drug), **rotenone** (a plant product commonly used as an insecticide), and **piericidin A** (an antibiotic) inhibit electron flow from the Fe-S centers of Complex I to ubiquinone and therefore block the overall process of oxidative phosphorylation.

Figure 21-29

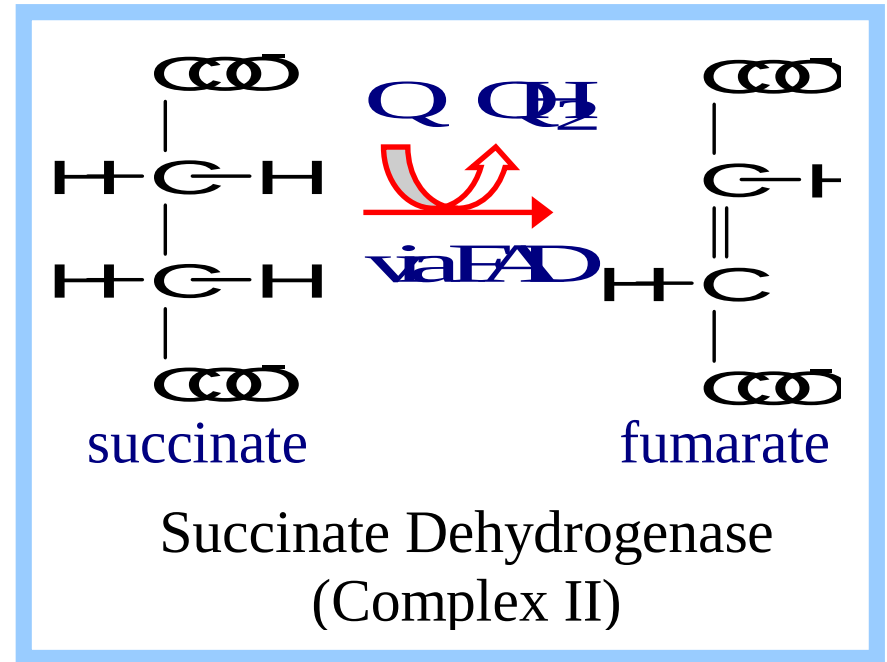


Complex II

Succinate Dehydrogenase

of the Krebs Cycle is also called **complex II** or Succinate-CoQ Reductase.

FAD is the initial e^- acceptor.



FAD is reduced to **FADH_2** during oxidation of succinate to fumarate.

FADH_2 is then reoxidized by transfer of electrons through a series of 3 iron-sulfur centers to CoQ, yielding **QH_2** .

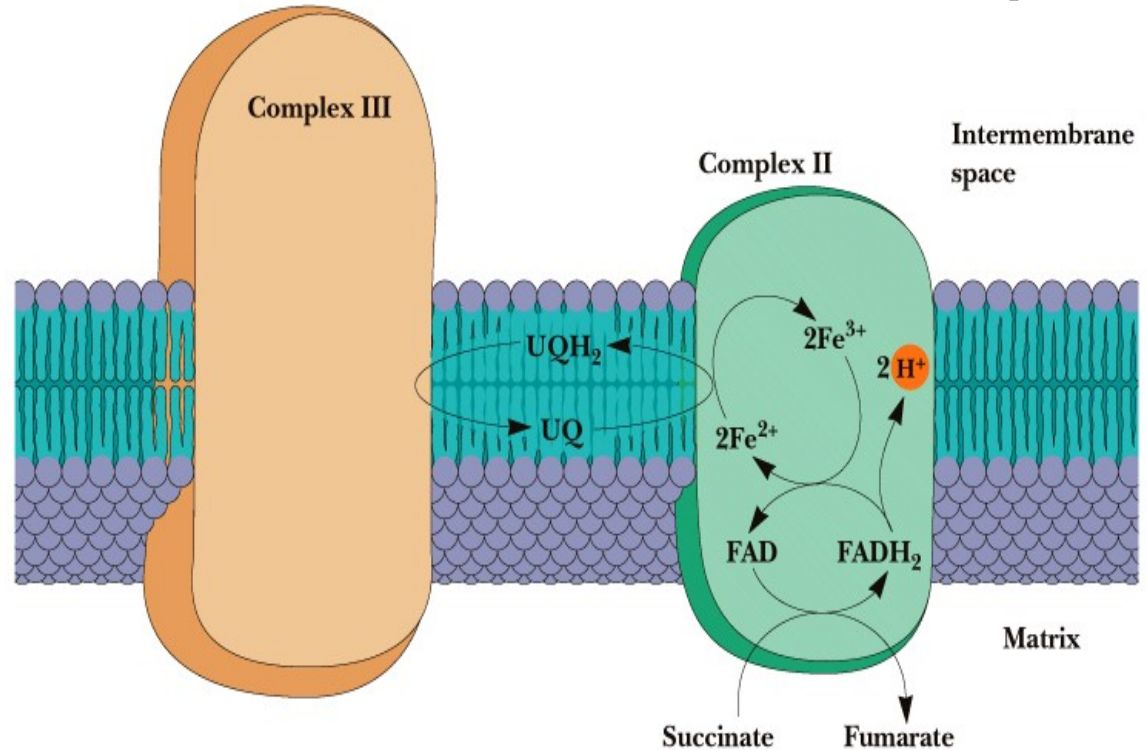
The **QH_2** product is then reoxidized via **complex III**, providing a pathway for transfer of electrons from succinate into the respiratory chain.

Complex II

Succinate-CoQ Reductase
or
Succinate dehydrogenase
(from TCA cycle!)

Garrett & Grisham: Biochemistry, 2/e
Figure 21.8

**H⁺ transport
does not occur
in this complex**

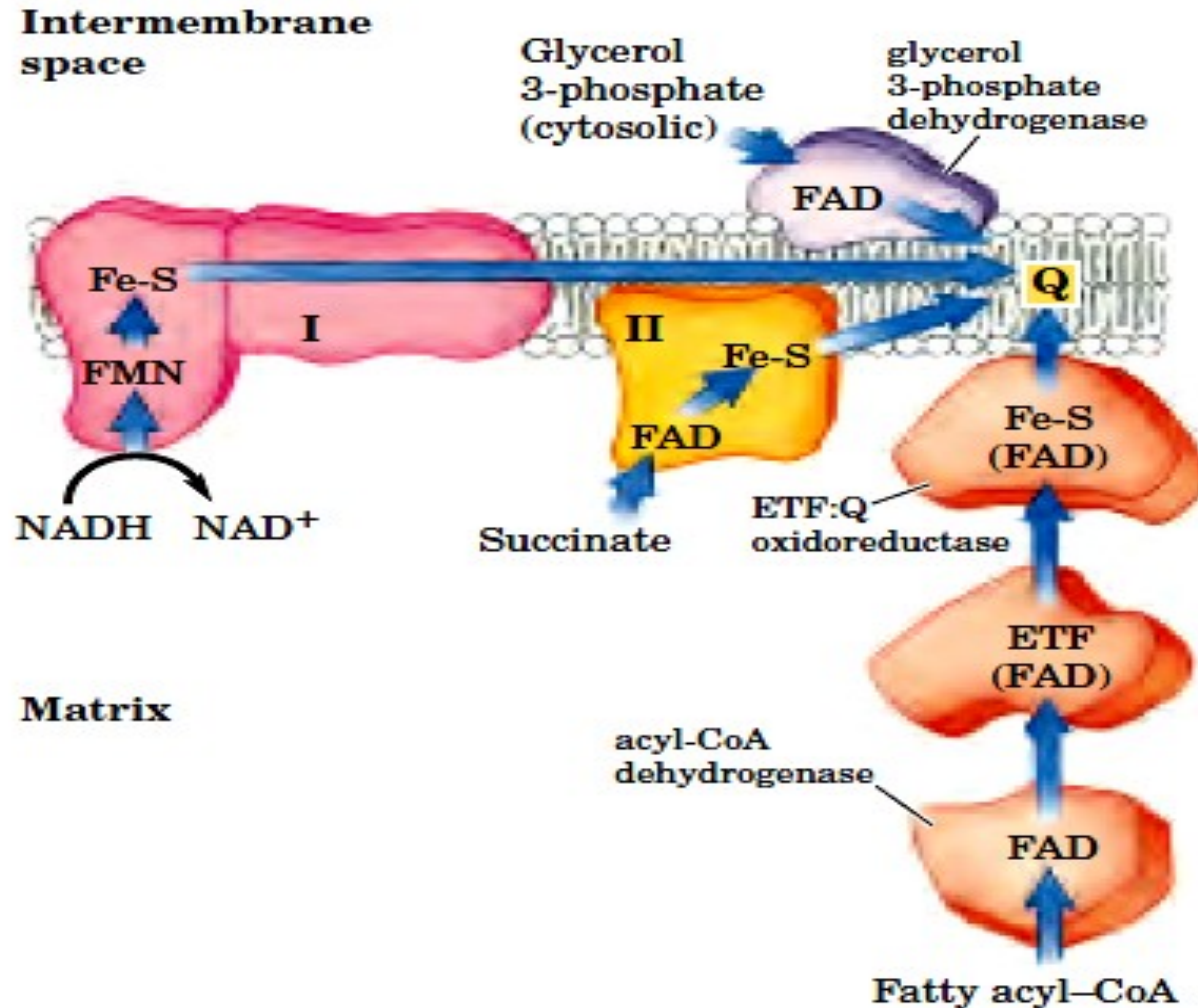


Saunders College Publishing

- Mass of 100 – 140 kD
- Composed of 4 subunits, including 2 Fe-S proteins
- Three types of Fe-S cluster: 4Fe-4S, 3Fe-4S, 2Fe-2S

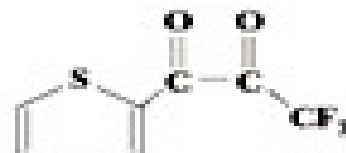
Path: Succinate → FADH₂ → 2Fe²⁺ → UQH₂

Path of electrons from NADH, succinate, fatty acyl-CoA, and glycerol 3-phosphate to ubiquinone.

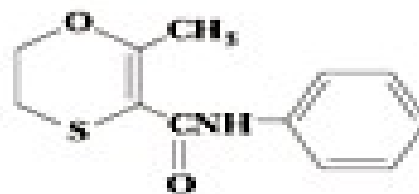


Inhibitors of complex II

2-Thenoyltrifluoroacetone &
carboxin block complex II



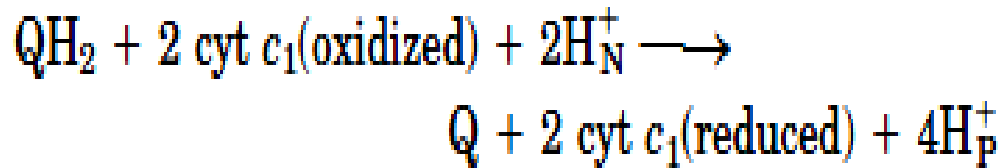
2-Thenoyltrifluoroacetone



Carboxin

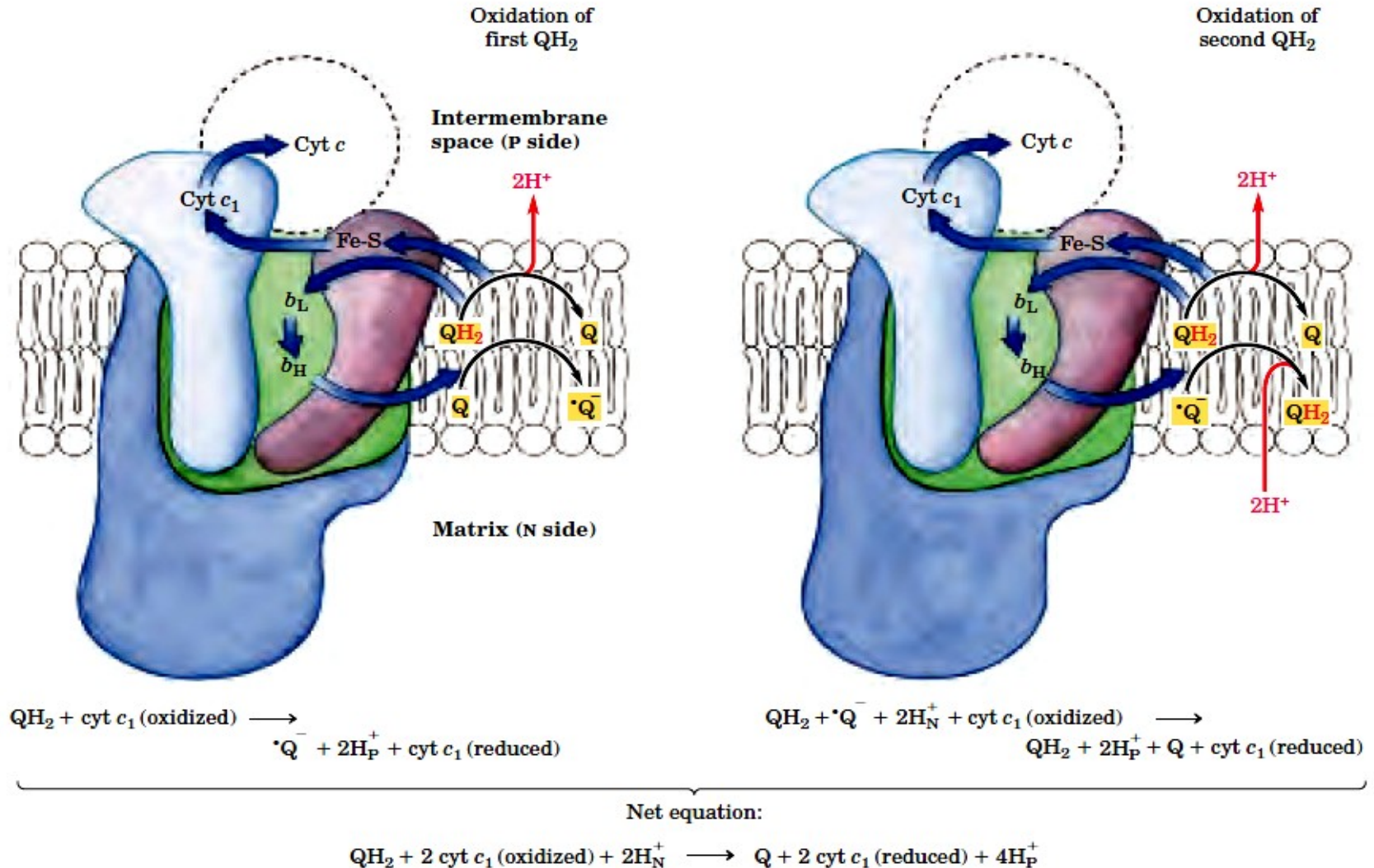
Complex III (Cytochrome Reductase)

- ❖ Accepts electrons from coenzyme **QH₂** that is generated by electron transfer in complexes I & II.
- ❖ Couples the transfer of electrons from ubiquinol (QH₂) to cytochrome *c* *with the vectorial* transport of protons from the matrix to the intermembrane space.



- ❖ Cytochrome *c*₁, a prosthetic group within complex III, reduces **cytochrome c**, which is the electron donor to complex IV.

The Q cycle.



Inhibitors of complex

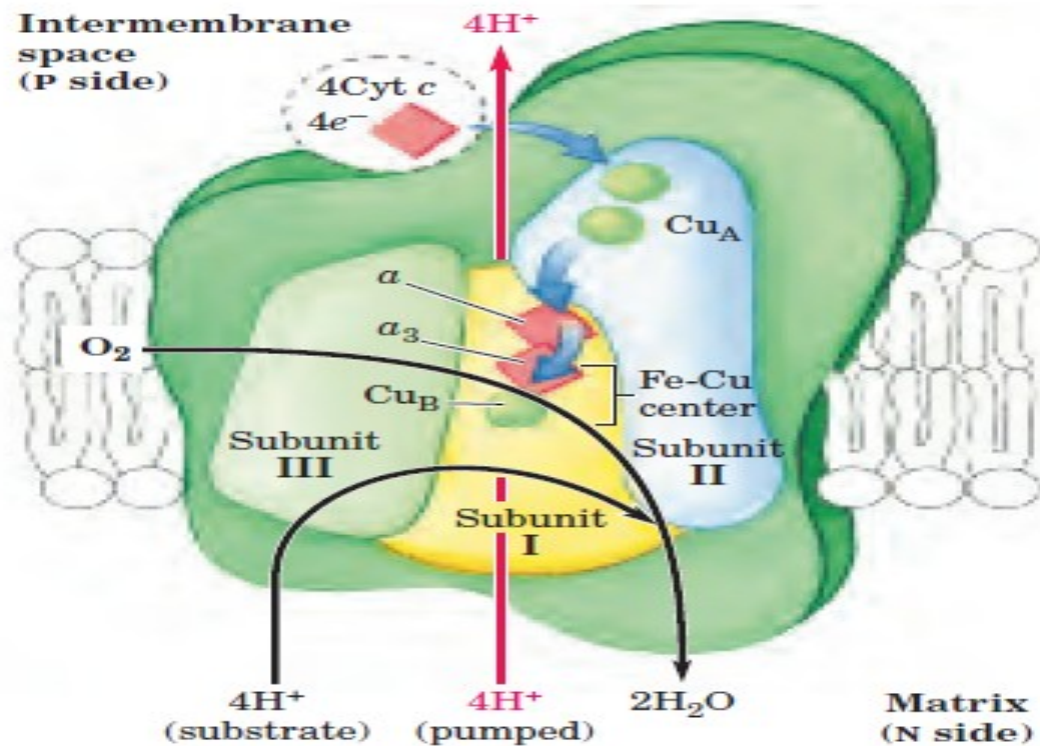
III

- ***Antimycin A:***

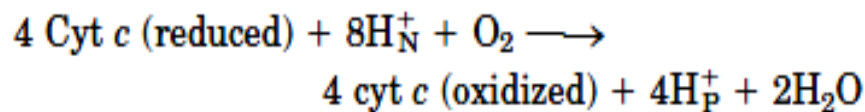
*An antibiotic blocks electron transport by inhibiting **cytochrome reductase**.*

- ***BAL (British anti lewisite):***

*Used as therapeutic agent in the cases of arsenic poisoning. It inhibits activity of **cytochrome reductase**.*



Cytochrome oxidase (complex IV) carries out the following irreversible reaction:



The four electrons are transferred into the complex one at a time from cytochrome c.

Inhibitors of complex

IV

- **Cyanide (CN)**

A powerful poison that inhibit **cytochrome oxidase** by combining with cytochrome a₃.

Cyanide may arise from cyanogenic substance.

- **Carbon monoxide (CO)**

It inhibits activity of cytochrome oxidase. (a pollutant).

- **3. Hydrogen sulfide (H₂S)**

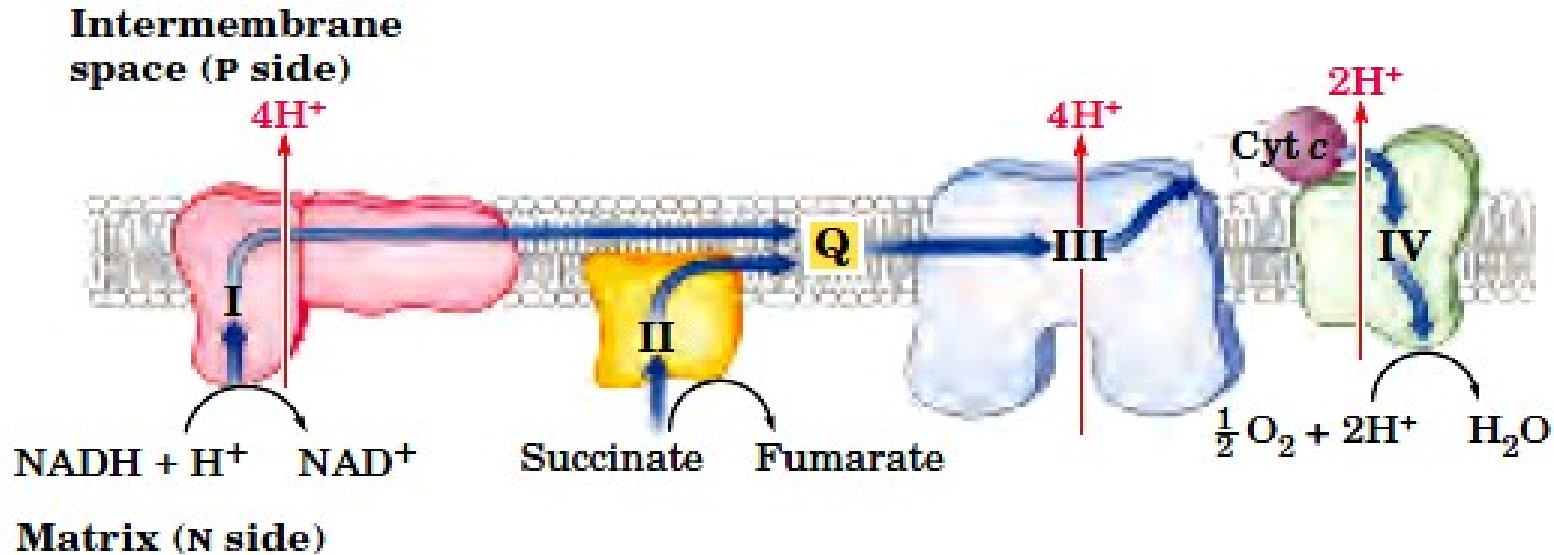
It inhibits cytochrome oxidase. H₂S toxicity occurs during oil drilling operations.

It is toxic as cyanide. It is a part of natural gas.

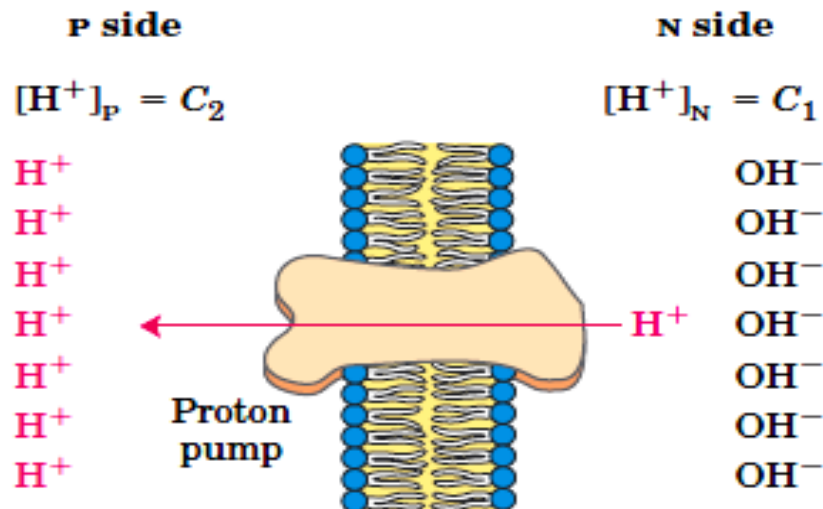
- **Azide:**

Sodium azide also inhibits cytochrome oxidase activity.

The Energy of Electron Transfer Is Efficiently Conserved in a Proton Gradient

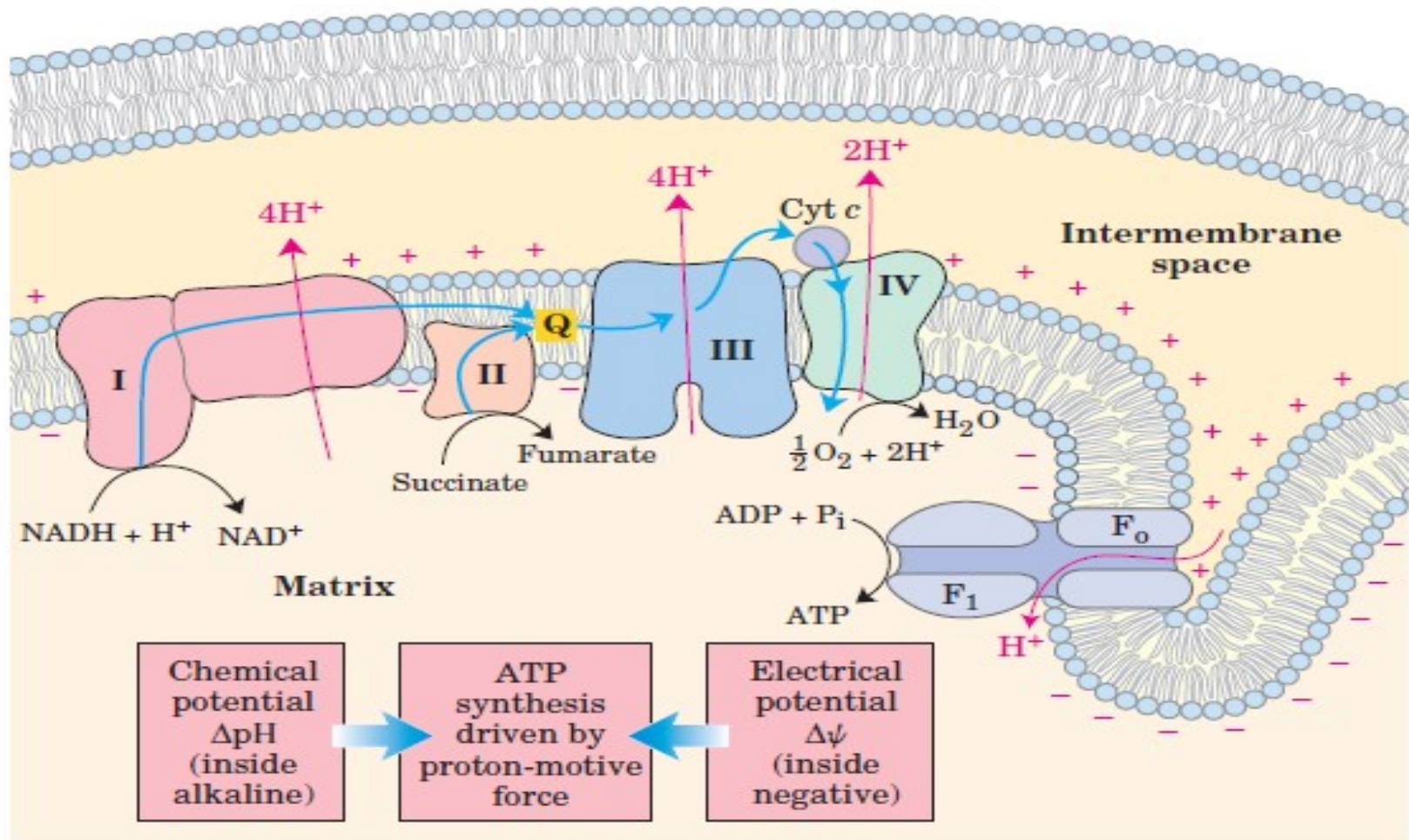


Proton gradient



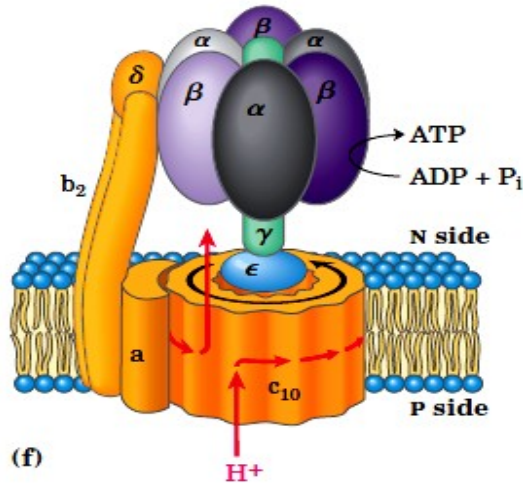
- Proton gradient created as electrons transferred to oxygen forming water
- $10 \text{H}^+ / \text{NADH}$
- $6 \text{H}^+ / \text{FADH}_2$

ATP synthesis



The Proton Gradient Drives ATP Synthesis. **How??**

F_1F_0 -ATP synthase of *E. Coli*.



F_1 portion

is composed of several subunits.

It contains three α , three β and one γ , δ and ϵ subunits, designated as $\alpha_3 \beta_3 \gamma_1 \delta_1 \epsilon_1$.

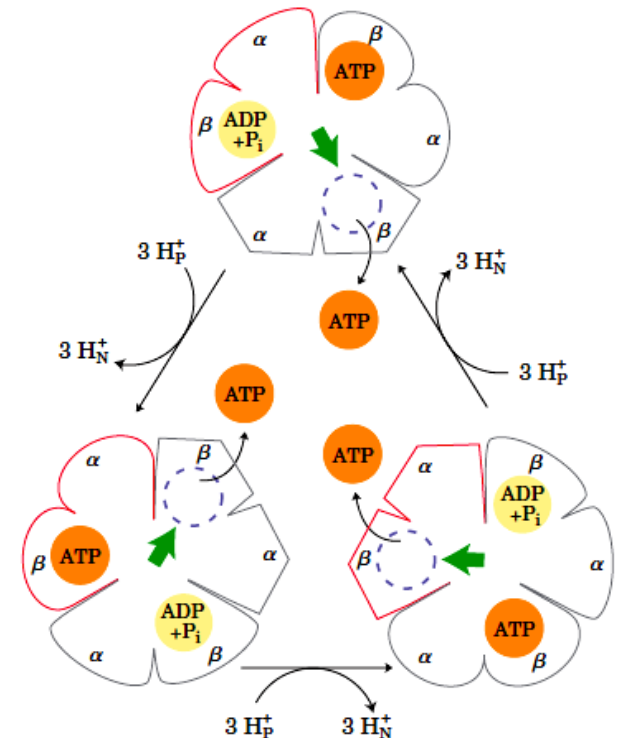
F_0 portion

is composed of one a, two b, and twelve c subunits.

Binding-change model for ATP synthase.

- Proposed by Paul Boyer.
- The F_1 complex has three nonequivalent adenine nucleotide-binding sites, one for each pair of α and β subunits.
- At any given moment, one of these sites is in the **β -ATP conformation** (which binds ATP tightly), a second is in the **β -ADP** (loose-binding) conformation, and a third is in the **β -empty** (very-loose-binding) conformation.
- proton-motive force causes rotation of the central shaft—the γ subunit comes into contact with each $\alpha\beta$ subunit pair in

Succession, producing cooperative conformational changes.



Generation of ATP

- Proton dependant ATP synthetase
 - Uses proton gradient to make ATP
 - Protons pumped through channel on enzyme
 - From intermembrane space into matrix
 - $\sim 4 \text{ H}^+ / \text{ATP}$
 - Called **chemiosmotic coupling theory**.

Other theories include: **Chemical coupling** theory & **Conformational coupling** theory.

Totals

NADH

$$10 \text{ H}^+ \times \underline{1 \text{ ATP}} = 2.5 \text{ ATP}$$

4 H^+

FADH₂

$$6 \text{ H}^+ \times \underline{1 \text{ ATP}} = 1.5 \text{ ATP}$$

4 H^+

Uncouplers of Oxidative phosphorylation

These compounds dissociates or uncouples oxidation in respiratory chain from phosphorylation.

So, the oxidation takes place without ATP synthesis.

Examples:

- 2, 4-dinitrophenol
- Dinitrocresol
- Salicylanilides
- Pentachlorophenol
- CCCP (Carbonylcyanide chloromethoxy phenyl hydrazone).
- FCCP (Carbanoyl cyanide p. trifluoromethoxy phenyl hydrazone).
- **Thermogenin**

Regulation of Oxidative phosphorylation.

Oxidative phosphorylation in the respiratory chain is subjected to regulation like any metabolic pathway.

The rate of respiration is generally limited by the availability of substrates such as: ADP, P_i , NADH, $FADH_2$ and O_2 .

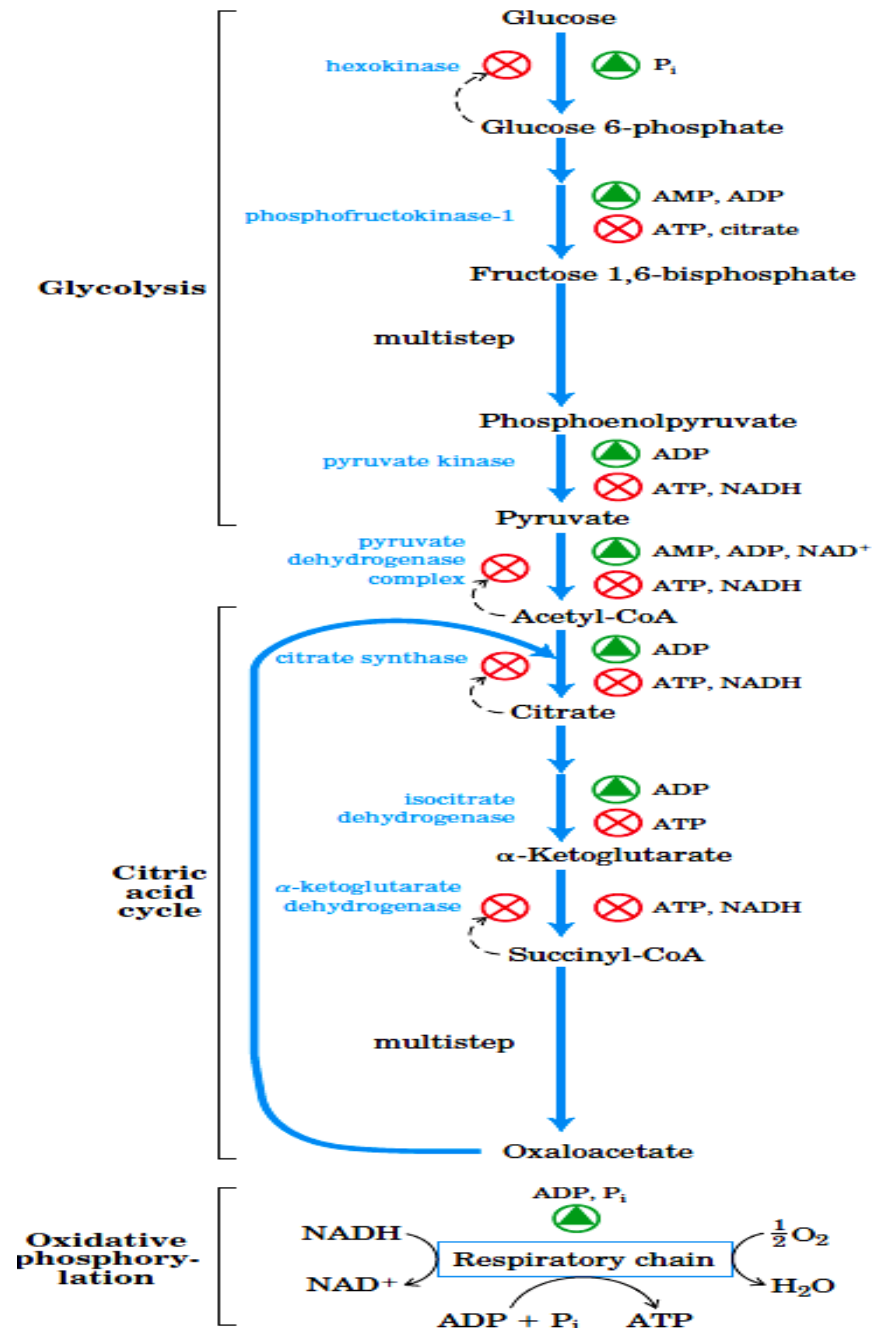


TABLE 19-4 Agents That Interfere with Oxidative Phosphorylation or Photophosphorylation

<i>Type of interference</i>	<i>Compound*</i>	<i>Target/mode of action</i>
Inhibition of electron transfer	Cyanide	Inhibit cytochrome oxidase
	Carbon monoxide	
	Antimycin A	Blocks electron transfer from cytochrome <i>b</i> to cytochrome <i>c</i> ₁
	Myxothiazol	Prevent electron transfer from Fe-S center to ubiquinone
	Rotenone	
	Amytal	
	Piericidin A	
Inhibition of ATP synthase	DCMU	Competes with Q _B for binding site in PSII
	Aurovertin	Inhibits F ₁
	Oligomycin	Inhibit F _o and CF _o
	Venturicidin	
Uncoupling of phosphorylation from electron transfer	DCCD	Blocks proton flow through F _o and CF _o
	FCCP	Hydrophobic proton carriers
	DNP	
	Valinomycin	K ⁺ ionophore
	Thermogenin	In brown fat, forms proton-conducting pores in inner mitochondrial membrane
Inhibition of ATP-ADP exchange	Atractyloside	Inhibits adenine nucleotide translocase

Oxidative phosphorylation

- This is the process by which ADP is phosphorylated with inorganic phosphate to generate ATP.
- NADH_2 has a central role in this process.
- Electrons are transferred down the respiratory chain from NADH_2 to oxygen. This is because NADH_2 is a strong electron donor while oxygen is a strong electron acceptor.